Predicting Follow-Up Year Emergency Department Utilization Among Non-Cancer U.S. Adults Using Pooled MEPS Longitudinal Panels

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Course: Integrated Capstone project.
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1. Introduction

Emergency department utilization is an important indicator of healthcare need, access, and care fragmentation. Individuals who use the emergency department frequently or unexpectedly may represent populations with unmet preventive care needs, chronic condition burden, or barriers to appropriate outpatient care. Because of this, predicting future ER utilization can support healthcare systems, payers, and population health programs in identifying groups that may benefit from earlier intervention and more proactive care management.

The goal of this XN project is to develop an interpretable predictive framework for follow-up year ER utilization among U.S. adults using the Medical Expenditure Panel Survey (MEPS). The project focuses specifically on adults aged 18 years and older and excludes individuals with evidence of cancer in order to maintain a consistent non-cancer cohort definition throughout the study. While earlier phases of the project focused on data loading, cohort construction, exploratory analysis, and baseline benchmarking, this reporting period emphasizes the final comparative modeling stage using pooled MEPS panels.

A major strength of the project is its longitudinal design. Rather than predicting a same-year outcome using variables collected at the same time, the model uses baseline-year information to predict ER utilization in the follow-up year. This structure helps preserve temporal ordering and avoids information leakage, making the modeling setup more defensible from both a methodological and an applied healthcare perspective.

2. Research Methods

2.1 Data Sources

This project uses pooled longitudinal data from five MEPS two-year panel files, along with the corresponding MEPS medical condition files needed for cancer exclusion. Each longitudinal file provides person-level demographic, socioeconomic, insurance, and utilization-related information across two survey years. The conditions files provide condition classification information that supports identification of respondents with cancer-related diagnoses.

Because each MEPS panel spans two calendar years, two conditions files were linked conceptually to each longitudinal panel in order to capture cancer exclusions across the full panel period. The resulting pooled dataset provided a broader and more stable analytic sample than any single panel alone.

2.2 Cohort Construction

The cohort was built in several stages. First, all five longitudinal panel files were loaded from fixed-width ASCII format using their official SPSS setup files. Second, all respondents younger

than 18 years were removed to restrict the analysis to adults. Third, conditions files were used to identify respondents with cancer-related clinical classification codes, and those individuals were excluded from the sample. This created a pooled adult non-cancer cohort suitable for predictive modeling.

Across the five panels, the final pooled cohort included 49,557 observations after adult restriction and cancer exclusion. This sample size provided a strong foundation for both descriptive analysis and predictive model comparison.

panel_file	n_start	n_adult	n_after_cancer_exclusion	outcome_prevalence
h114	16533	11257	10506	0.1361
h122	12440	8557	7833	0.1252
h130	18287	12367	11438	0.1326
h139	16221	11311	10440	0.1217
h148	14541	10089	9340	0.133

2.3 Outcome Definition

The primary response variable for this phase was follow-up year emergency department utilization, defined as a binary indicator of whether an individual had any ER use in Year 2. Respondents with one or more Year-2 emergency department visits were coded as 1, while respondents with no ER use were coded as 0.

This outcome is clinically and operationally relevant because ER utilization is often associated with unmet care needs, chronic disease burden, and elevated healthcare risk. It also supports practical risk stratification use cases in population health management.

2.4 Predictor Construction

Predictors were constructed using baseline-year information only. Baseline features included age group, sex, insurance status, poverty category, region, and panel identifier. In the enhanced model, prior-year total expenditure and a log-transformed version of that expenditure variable were added to capture healthcare cost burden and utilization intensity from the baseline year.

Age was grouped into bands to improve interpretability, and log transformation was applied to prior-year expenditures to reduce skewness and improve model stability. Missing negative-coded MEPS responses were standardized as missing values during preprocessing.

2.5 Modeling Strategy

Three predictive models were evaluated:

1. **Baseline Logistic Regression Model**
   Used only baseline demographic and access-related categorical predictors. This model served as the first interpretable benchmark.

2. **Enhanced Logistic Regression Model**
   Added prior-year expenditure variables to the baseline model in order to test whether expenditure history improved follow-up year ER prediction.

3. **Boosted Tree Benchmark Model**
   Served as a nonlinear comparison model to evaluate whether a more flexible model would provide stronger predictive performance than the logistic models.

Model performance was evaluated using ROC-AUC, precision-recall analysis, confusion matrices, threshold sweep analysis, and calibration-style summaries.

3. Analysis Results and Discussion

3.1 Cohort Overview

The pooled MEPS cohort provided a large, longitudinally structured adult sample for analysis. After cancer exclusion, the final modeling dataset contained 49,557 observations. The outcome prevalence for follow-up year ER utilization was approximately 13.0%, indicating a moderately imbalanced but still workable binary classification problem.

panel_file	n_start	n_adult	n_after_cancer_exclusion	outcome_prevalence
h114	16533	11257	10506	0.1361
h122	12440	8557	7833	0.1252
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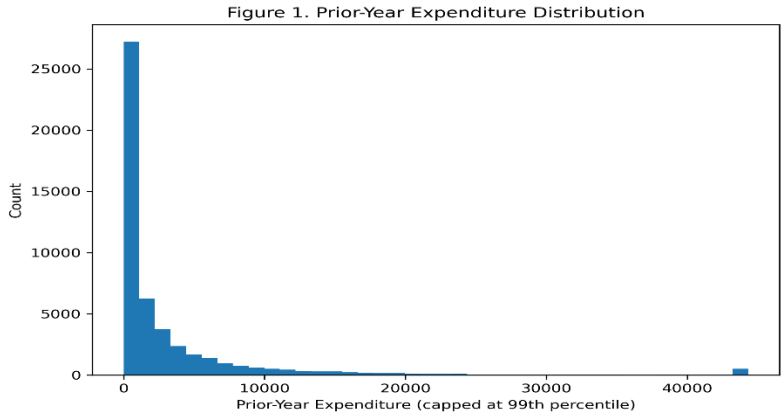
h139	16221	11311	10440	0.1217
h148	14541	10089	9340	0.133

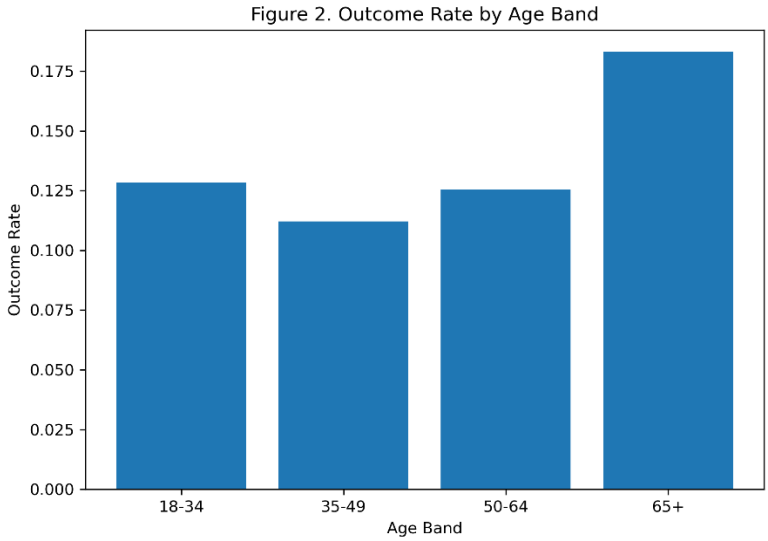
This class distribution is realistic for healthcare utilization prediction, where most individuals do not experience the event of interest, but a smaller subgroup does. The imbalance also explains why metrics beyond accuracy, such as ROC-AUC, PR-AUC, recall, and threshold-based tradeoffs, are necessary for interpreting model performance.

3.2 Descriptive Patterns

Prior-year expenditures showed the expected right-skewed pattern commonly seen in healthcare cost data, where a relatively small proportion of individuals account for a much larger share of total spending. This supported the decision to include a log-transformed version of prior-year expenditures in the enhanced model.



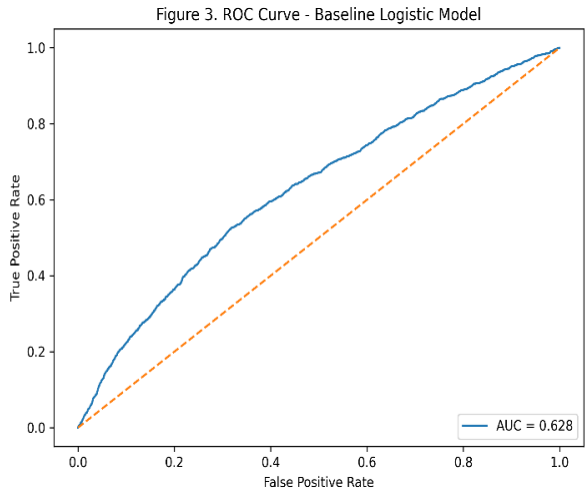
Outcome rates also varied across age groups. Older age bands generally showed higher follow-up year ER utilization than younger groups, which is consistent with broader healthcare utilization patterns and reinforces the value of age as an important baseline predictor.

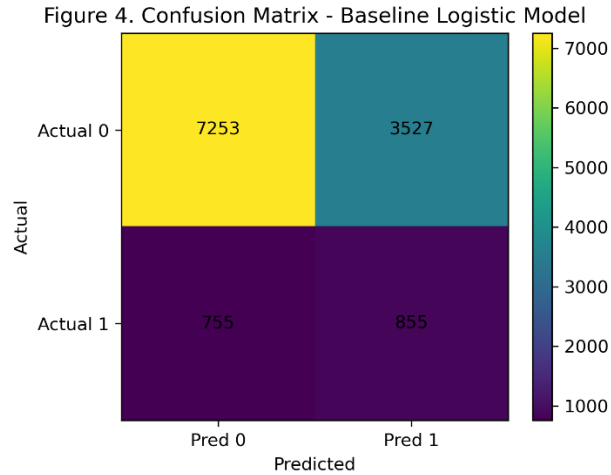


3.3 Baseline Logistic Regression

The baseline logistic regression model used only baseline demographic and access-related categorical predictors. On the test set, this model achieved an ROC-AUC of **0.6285**, indicating modest discriminatory ability. This suggests that baseline demographic and access-related

features alone provide some predictive signal, but not enough to fully distinguish future ER users from non-users.

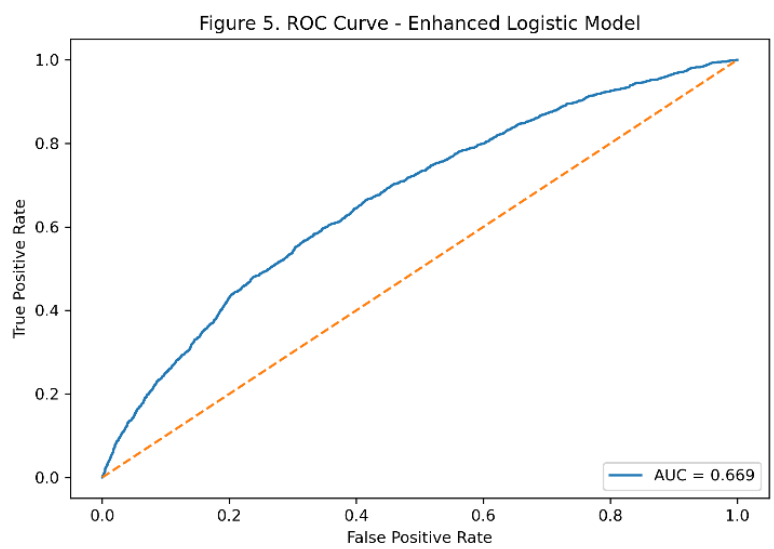


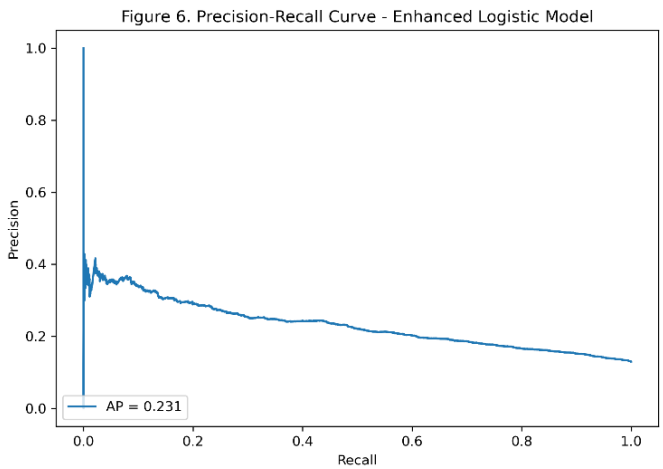


At the default threshold of 0.50, the baseline model showed relatively high precision for the non-ER class but much lower precision for the positive class. However, it still captured a meaningful portion of future ER users, which makes it a useful benchmark for comparison.

3.4 Enhanced Logistic Regression

The enhanced logistic regression model added prior-year spending information, specifically total prior-year expenditures and log-transformed prior-year expenditures. This model improved performance meaningfully, achieving an ROC-AUC of **0.6688** and a PR-AUC of **0.2311**.





This performance gain shows that prior healthcare spending history adds useful predictive information beyond demographics and access variables alone. In practical terms, baseline expenditure appears to capture part of the healthcare intensity signal that is relevant to future emergency department use.

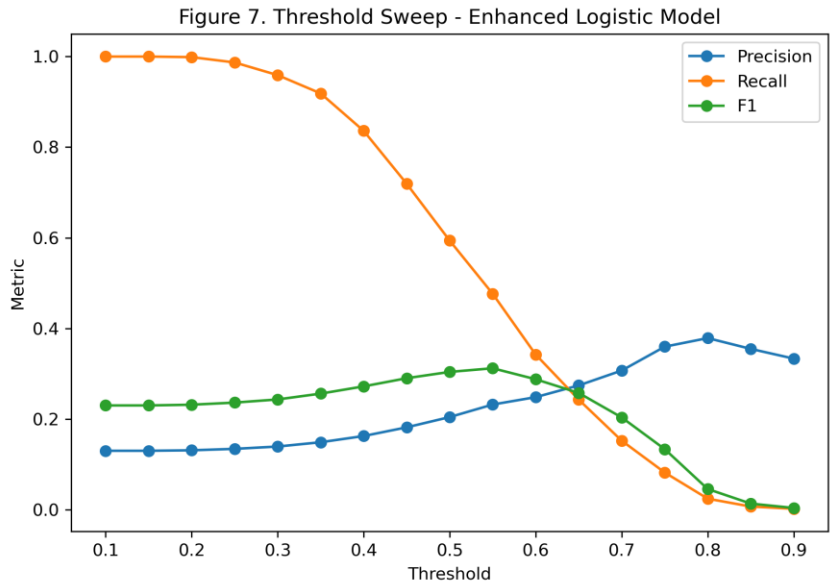
At the same time, the improvement remained moderate rather than dramatic. This suggests that while expenditure history matters, ER utilization is still influenced by multiple underlying factors not fully captured by the available baseline variables.

3.5 Threshold Sweep Analysis

A threshold sweep was conducted for the enhanced logistic model in order to evaluate how precision, recall, and F1-score changed under different classification thresholds. This analysis is particularly important in applied healthcare settings because the “best” threshold depends on how the model is intended to be used.

For example, a lower threshold may be preferred when the goal is to identify more potentially high-risk individuals for broader outreach, while a higher threshold may be preferred when intervention capacity is limited and precision is more important.

column	missing_rate
AGE_BAND	0.00
SEX_BASE	0.00
INS_BASE	0.00
POV_BASE	0.00
REGION_BASE	0.01
FILE_CODE	0.00
TOTEXPY1_FINAL	0.00
LOG_TOTEXPY1	0.00
ER_ANY_Y2	0.00



The threshold analysis improves the operational value of the model by moving beyond a single default threshold and showing how decision-makers can tailor the model to different use cases.

3.6 Boosted Tree Benchmark

To determine whether a nonlinear model could improve performance further, a boosted tree benchmark was trained using the same predictor set as the enhanced logistic model. The boosted tree achieved an ROC-AUC of **0.6689**, a PR-AUC of **0.2345**, and a Brier score of **0.1079**.

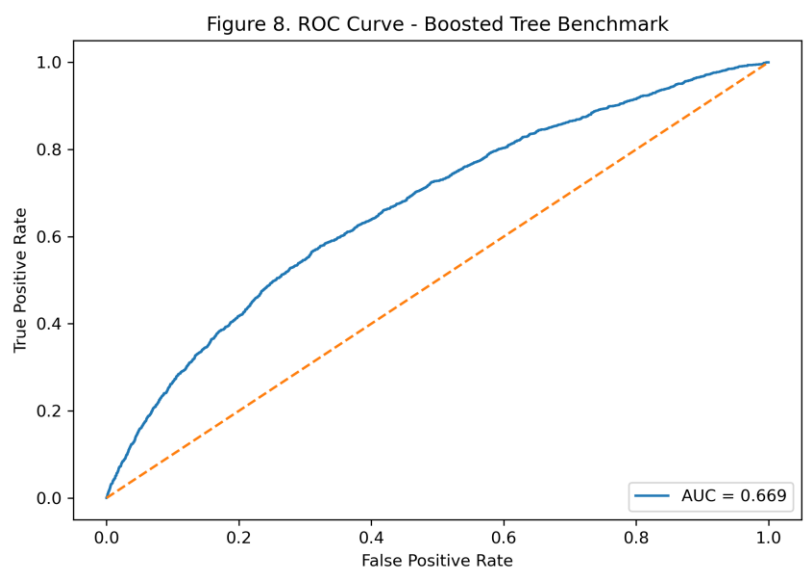
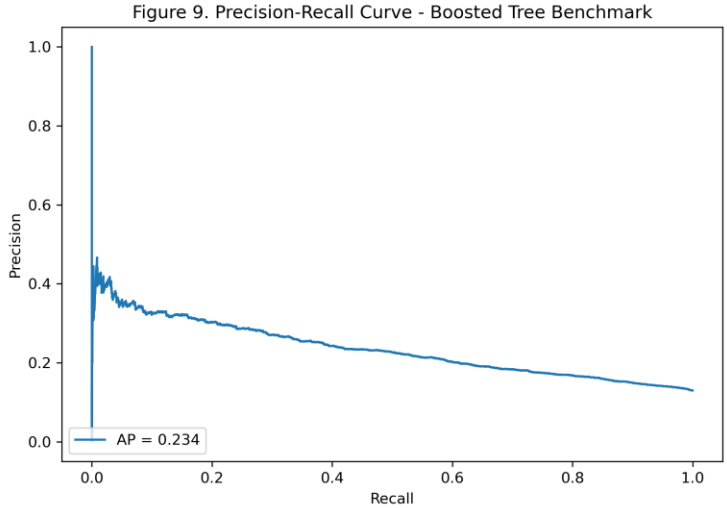


Figure 9. Precision-Recall Curve - Boosted Tree Benchmark

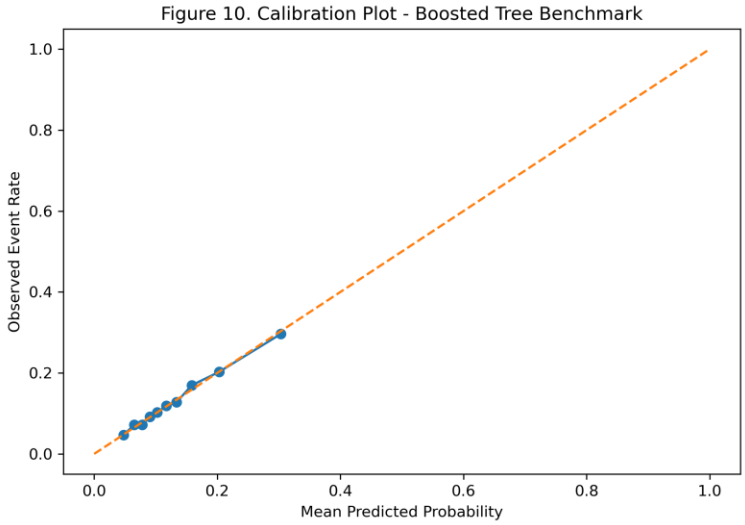


The results show that the boosted tree benchmark performed only marginally better than the enhanced logistic model. The difference in ROC-AUC was negligible, and the improvement in PR-AUC was also very small. This is an important finding because it suggests that the simpler enhanced logistic model captures most of the useful predictive signal available in the chosen feature set.

A calibration-style summary was also produced for the boosted tree model to assess the alignment between predicted probabilities and observed event rates.

AGE_BAND	outcome_rate
18-34	0.128256635
35-49	0.112126631
50-64	0.125566795
65+	0.183023468

Figure 10. Calibration Plot - Boosted Tree Benchmark

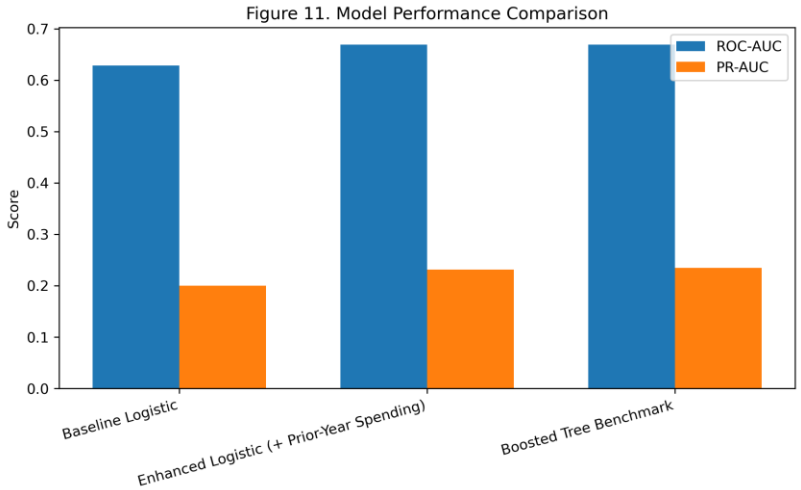


This analysis adds interpretive depth by showing not only how well the model ranks individuals by risk, but also how closely predicted risks correspond to observed outcomes across score bands.

3.7 Model Comparison

The final comparison across models highlights the tradeoff between simplicity and predictive performance. The baseline logistic model provided the most interpretable starting point but had the weakest discrimination. The enhanced logistic model improved performance noticeably through the addition of prior-year expenditure. The boosted tree benchmark delivered only a very small improvement beyond the enhanced logistic model.

threshold	precision	recall	f1	TP	FP	FN	TN
0.1	0.129943503	1	0.23	1610	10780	0	0
0.15	0.129943503	1	0.23	1610	10780	0	0
0.2	0.130997963	0.998757764	0.231616853	1608	10667	2	113
0.25	0.134194747	0.986956522	0.236264962	1589	10252	21	528
0.3	0.139299892	0.959006211	0.243264534	1544	9540	66	1240
0.35	0.148792757	0.91863354	0.256103896	1479	8461	131	2319
0.4	0.162481893	0.836024845	0.272084091	1346	6938	264	3842
0.45	0.181704064	0.719254658	0.290116498	1158	5215	452	5565
0.5	0.204317162	0.59378882	0.304022897	956	3723	654	7057
0.55	0.23200242	0.476397516	0.312042311	767	2539	843	8241
0.6	0.24831005	0.342236025	0.287803604	551	1668	1059	9112
0.65	0.274193548	0.242857143	0.257575758	391	1035	1219	9745
0.7	0.307017544	0.152173913	0.203488372	245	553	1365	10227
0.75	0.359673025	0.081987578	0.13353566	132	235	1478	10545
0.8	0.378640777	0.024223602	0.045534151	39	64	1571	10716
0.85	0.35483871	0.006832298	0.013406459	11	20	1599	10760
0.9	0.333333333	0.001863354	0.003705991	3	6	1607	10774



Model	ROC_AUC	PR_AUC
Baseline Logistic	0.62849932	0.199640443
Enhanced Logistic (+ Prior-Year Spending)	0.668849664	0.231134308
Boosted Tree Benchmark	0.668861937	0.234457666

From a practical perspective, this makes the enhanced logistic model especially attractive. It offers substantially better performance than the baseline benchmark, remains interpretable, and performs nearly as well as the more complex boosted tree model. For healthcare decision-making contexts where transparency matters, this balance is valuable.

4. Conclusion

This biweekly period represented the final major modeling stage before capstone project finalization. Using a pooled adult non-cancer MEPS cohort, I compared a baseline logistic regression model, an enhanced logistic regression model with prior-year spending, and a boosted tree benchmark for predicting follow-up year emergency department utilization.

The findings showed that baseline demographic and access variables alone provide only modest predictive ability, but the addition of prior-year expenditure improves performance meaningfully. Although the boosted tree benchmark performed slightly better than the enhanced logistic model, the difference was minimal. As a result, the enhanced logistic model appears to offer the best overall balance between interpretability and predictive strength.

These results strengthen the project's final analytical framework and provide a clear direction for finalization: emphasize a leakage-safe, interpretable predictive design while supporting results with threshold-based and calibration-style evaluations that make the findings more useful for real-world healthcare applications.

5. References

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